

EB-JEPA: Energy-Based Joint Embedding Predictive Architecture for Self-Supervised EEG Representation Learning

A Technical Research Report

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Abstract

We present EB-JEPA, an Energy-Based Joint Embedding Predictive Architecture for self-supervised learning of EEG representations applied to clinical abnormality detection. Starting from a single EEG window, two independently corrupted views are produced through different augmentation pathways and mapped to the same latent space by a shared encoder trained without labels. Four encoder families are evaluated on the TUH Abnormal EEG Corpus (TUAB v3.0.1 [15], 2717 training and 276 evaluation recordings): a temporal Conv1D network (0.4M parameters), a patch Transformer inspired by LaBraM [6] (1.2M), an FFT-tokenised Transformer inspired by BIOT [22] (1.2M), and a hierarchical channel-pool Transformer inspired by EEGPT [20] (0.8M). We further introduce NaiveJEPA, a temporal-split predictive variant requiring no anti-collapse regulariser, evaluated on both TUAB and the 6-class TUH EEG Events corpus (TUEV v2.0.0) to assess cross-task transfer.

Our best frozen encoder (Conv1D + corruption augmentation + SIGReg) reaches **0.775 BACC per-window** and **0.825 BACC per-recording** on TUAB, matching published ContraWR and approaching LaBraM-Base (0.814). The central contribution is *systematic diagnosis*: we document three failure modes — a VICReg coefficient bug ($\lambda=1$ vs. paper-correct $\lambda=25$), an architectural conflict between EEGPT’s channel-pooling and channel-drop augmentation, and a per-window vs. per-recording evaluation gap (≈ 5 pp) that initially caused an inflated performance claim. A new hyperparameter sweep with fixed VICReg reveals that the spectral auxiliary loss (previously best-in-class at 0.836 per-recording with the buggy coefficients) provides no independent benefit over correctly-tuned VICReg alone (0.819), resolving a key open question.

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Introduction

Motivation. Clinical EEG abnormality detection is a high-stakes task: a board-certified neurologist must review each recording to label it as normal or abnormal, a process that is time-consuming and subject to inter-rater variability. Self-supervised learning (SSL) offers the prospect of learning useful representations from the vast majority of unlabelled clinical EEG recordings, requiring only a small labelled probe set. The Energy-Based Joint Embedding Predictive Architecture (EB-JEPA [18]) provides a unified framework in which the SSL objective is expressed as an energy function to be minimised, with anti-collapse regularisers preventing the trivial constant solution.

Report structure.

Section 1: State of the Art. We first define three distinct *scoring styles* used across the EEG literature, then provide the complete provenance table of published results on TUAB, verified against primary PDFs.

Section 2: EB-JEPA Model. We describe the two-view energy concept, the dataset with its SSL/probe splits, and the full experimental results from all encoder $\times$ loss combinations tested.

Section 3: Per-Model Mathematical Descriptions. For every model, we provide the complete energy function with formal component definitions, architecture details, the downstream MLP classifier, and the exact training/evaluation protocol.

Section 4: Combined Best Results. We consolidate the best per-recording numbers from literature and this work.

Section 5: Conclusion and Future Work. Open scientific questions, methodological lessons, and prioritised future experiments.

Research question. Can a small encoder (≤ 1.2 M parameters) pre-trained without labels on 2717 EEG recordings learn representations that generalise to held-out patients and match supervised models trained with labels?

1 State of the Art on TUAB EEG Abnormality Detection

1.1 Scoring Styles

Published TUAB results use heterogeneous evaluation conventions. We define four *scoring styles* precisely:

(S1) Per-window (WIN). The evaluation set is segmented into all non-overlapping 10-second windows. Each window is independently classified and inherits the binary label of its parent recording. BACC is computed over all individual window predictions. For TUAB eval this yields ≈ 409 k test windows from 2339 recordings (the larger published eval split) or ≈ 400 k from 276 recordings (our official TUAB v3.0.1 split). This is the convention of BIOT [22], LaBraM [6], ContraWR [21], and most published SSL models. The test samples are *not independent*: windows from the same recording share the label and much of the signal, inflating effective sample size.

(S2) Per-recording via window embedding pooling (REC-WIN). A fixed number N of evenly-spaced windows are extracted per recording and each is encoded separately; the N embeddings are mean-pooled into one vector. One prediction is made per recording. This is the default mode of our framework with $N=16$. Noise averaging over 16 windows systematically raises BACC by ≈ 5 pp over WIN for the same encoder and probe.

(S3) Per-recording via majority vote (REC-VOTE). Each window is independently classified; the recording label is the majority class. This style appears in some clinical pipelines but not in the SSL literature surveyed here.

(S4) Per-recording from the full recording (REC-FULL). The entire recording (variable length, up to 120 min) is fed to a model capable of handling variable-length sequences (SSM, hierarchical model). Used by FEMBA [17] in its fine-tuned evaluation.

Critical warning. Comparing REC-WIN or REC-FULL numbers to WIN numbers as if they were the same quantity produces a spurious ≈ 5 pp advantage for the per-recording model. Section 5 describes how this gap caused an initial erroneous claim.

1.2 State-of-the-Art Provenance Table

Table 1 compiles every TUAB result we could verify against a primary PDF. Every literature number is backed by a specific table in a specific paper. Entries in yellow come from benchmark papers (EEG-Bench [8], FEMBA [17]) rather than original method papers. Entries in green are our own results.

Key observations.

1. At WIN protocol, the state of the art is LaBraM-Huge (0.826) requiring 369M parameters pretrained on massive external data. Among TUAB-only methods: LaBraM-Base (0.814) and BIOT-pretrained (0.802) are the targets.
2. Our best WIN result (0.775, Conv1D + SIGReg + corrupt) matches ContraWR (0.775) and is 3.9 pp below LaBraM-Base, with only 0.4M parameters and 20 training epochs.
3. The REC-WIN vs WIN gap is ≈ 5 pp for the same encoder: our best REC-WIN (0.836, buggy VICReg) vs WIN (0.765 same run) gives a 7.1 pp gap. This gap was initially used to claim we “beat LaBraM,” which is incorrect.
4. The spectral audit [3] finds that many published methods rely on the aperiodic ($1/f$) spectral slope for ΔBACC 0.07–0.13; whether EB-JEPA representations share this shortcut is an open question.

2 EB-JEPA Model

2.1 EB-JEPA Two-View Energy Concept

EB-JEPA is a two-view energy-based SSL framework for EEG, inspired by the Joint Embedding Predictive Architecture paradigm [1, 13]. The key idea is:

Starting from one EEG window, two corrupted views are created via independent augmentation pathways (a) and (b), and an encoder (the “world model”) is trained to map both views to the same point in latent space.

Formal setup. Given a 10-second EEG window $\mathbf{x} \in \mathbb{R}^{C \times T}$ with $C=19$ channels and $T=2000$ samples at 200 Hz:

- **Pathway (a):** $\tilde{\mathbf{x}}_a = \mathcal{A}_a(\mathbf{x})$ — apply augmentation stack $\mathcal{A}_a$ with independently sampled parameters.
- **Pathway (b):** $\tilde{\mathbf{x}}_b = \mathcal{A}_b(\mathbf{x})$ — apply the same augmentation family with a different random seed.

Table 1: Provenance table: verified results on TUAB EEG abnormality detection. WIN = per-window BACC; REC = per-recording (pooled). FT = fine-tuned; Frozen = frozen encoder + probe. All WIN numbers use the standard BIOT/LaBraM/FEMBA/EEGPT convention (409k windows). Our own runs use the official TUAB v3.0.1 eval split (276 recordings). [†]Accuracy (not BACC); TUAB near-balanced (54%/46%), so $\text{acc} \approx \text{BACC}$ within 2 pp. [‡]Protocol unclear; BACC derived from reported sensitivity + specificity. Checks (✓) indicate the value appears verbatim in the cited paper.

Style	Model	BACC	AUROC	Params	Source	Mode	Ver.
Foundation models (large external pretraining)							
WIN	LaBraM-Huge [6]	0.8258	0.9162	369M	Jiang 2024, Tbl 1	FT	✓
WIN	LaBraM-Large [6]	0.8226	0.9127	46M	Jiang 2024, Tbl 1	FT	✓
WIN	FEMBA-Huge [17]	0.8182	0.8921	386M	Tegon 2025, Tbl II	FT	✓
WIN	FEMBA-Large [17]	0.8147	0.8856	77.8M	Tegon 2025, Tbl II	FT	✓
WIN	LaBraM-Base [6]	0.8140	0.9022	5.8M	Jiang 2024, Tbl 1	FT	✓
WIN	FEMBA-Base [17]	0.8105	0.8829	47.7M	Tegon 2025, Tbl II	FT	✓
WIN	EEG2Rep [17]	0.8052	–	–	Tegon 2025, Tbl II	FT	✓
Self-supervised models (TUAB-only or limited external pretraining)							
WIN	BIOT [22] (pretrained, PREST+SHHS)	0.8019	0.8739	3.2M	Yang 2023, Tbl 4	Frozen	✓
WIN	AFTA	0.8002	0.8848	–	Ref. not public	FT	✓
WIN	EEGPT-25M [20]	0.7983	0.8718	25M	Wang 2024, Tbl 3+9	Frozen	✓
WIN	ST-Transformer [10]	0.7966	0.8707	3.5M	LaBraM Tbl 1 / BIOT Tbl 4	FT	✓
WIN	BIOT [22] (6 EEG datasets)	0.7959	0.8815	3.2M	Yang 2023, Tbl 4	Frozen	✓
WIN	EEGPT-Tiny [20]	0.7959	0.8716	4.7M	Wang 2024, Tbl 3+9	Frozen	✓
WIN	BIOT [22] (vanilla supervised)	0.7925	0.8691	3.2M	Yang 2023, Tbl 4	FT	✓
WIN	SPaRCNet [7]	0.7896	0.8676	0.79M	LaBraM Tbl 1 / BIOT Tbl 4	FT	✓
WIN	FFCL [14]	0.7848	0.8569	2.4M	Jiang 2024, Tbl 1	FT	✓
WIN	CNN-Transformer	0.7777	0.8461	3.2M	Jiang 2024, Tbl 1	FT	✓
WIN	ContraWR [21]	0.7746	0.8456	1.6M	Jiang 2024, Tbl 1	Frozen	✓
WIN	BENDR [11] (FEMBA re-run)	0.7696	0.8397	0.39M	Tegon 2025, Tbl II	FT	✓
Supervised CNN baselines (Schirrmeister 2017 / spectral audit)							
WIN	Deep4Net [‡]	0.846	–	–	Schirrmeister 2017, Tbl II	FT	✓
WIN	ShallowConvNet [‡]	0.839	–	44k	Schirrmeister 2017, Tbl II	FT	✓
WIN	EEGNet [12]	0.804	–	9k	Spectral Audit [3], Tbl 1	FT	✓
WIN	Deep4Net [16]	0.816	–	–	Spectral Audit [3], Tbl 1	FT	✓
EB-JEPA (this work) — WIN (per-window)							
WIN	Conv1D + VICReg* (base)	0.756	–	0.4M	This work	Frozen	
WIN	Conv1D + VICReg* + corrupt	0.770	–	0.4M	This work	Frozen	
WIN	Conv1D + SIGReg + corrupt	0.775	–	0.4M	This work	Frozen	
WIN	Conv1D + VICReg* + spec 0.1	0.765	–	0.4M	This work	Frozen	
EB-JEPA (this work) — REC-WIN (N=16, per-recording)							
REC-WIN	Conv1D + VICReg (fixed)	0.819	0.900	0.4M	This work	Frozen	
REC-WIN	Conv1D + SIGReg + corrupt	0.825	0.904	0.4M	This work	Frozen	
REC-WIN	Conv1D + VICReg* + spec 0.1	0.836	0.887	0.4M	This work (buggy)	Frozen	
REC-WIN	Conv1D + VICReg* + corrupt (FT)	0.812	0.908	0.4M	This work	FT	

VICReg* = buggy `inv_coeff=1`; VICReg (fixed) = paper-correct `inv_coeff=25`.

Both views are corrupted versions of the same brain state. A shared encoder f_θ (the world model) and projector g_ϕ map each view to a latent vector:

$$\mathbf{z}_a = g_\phi(f_\theta(\tilde{\mathbf{x}}_a)), \quad \mathbf{z}_b = g_\phi(f_\theta(\tilde{\mathbf{x}}_b)), \quad \mathbf{z}_a, \mathbf{z}_b \in \mathbb{R}^{d_p}, \quad d_p=1024. \quad (1)$$

The energy function $E(\mathbf{z}_a, \mathbf{z}_b)$ is minimised during training:

$$E = \underbrace{\|\mathbf{z}_a - \mathbf{z}_b\|^2}_{\text{invariance}} + w_V \underbrace{L_V(\mathbf{z}_a, \mathbf{z}_b)}_{\text{anti-collapse (variance)}} + w_C \underbrace{L_C(\mathbf{z}_a, \mathbf{z}_b)}_{\text{decorrelation (covariance)}}. \quad (2)$$

Without the anti-collapse terms, the trivial global minimum $\mathbf{z}_a = \mathbf{z}_b = \mathbf{0}$ satisfies $E=0$ while encoding no information. VICReg [2] and SIGReg prevent this.

The encoder acts as a **world model**: it abstracts away the specific corruption and retains the clinically relevant brain state. Pathological brains manifest statistically different signal statistics across all augmentation families; the encoder is forced to represent what is preserved across corruptions, which correlates with pathology.

2.2 EEG Dataset: SSL Training vs. Probe Evaluation Splits

Dataset. All experiments use the **TUH Abnormal EEG Corpus (TUAB) v3.0.1** [15]. 19-channel scalp EEG in the 10-20 system, binary normal/abnormal labels assigned by a board-certified neurologist per recording.

Offline preprocessing (provided by dataset). Applied once to the raw files before any experiment:

1. **Channel selection:** 19 standard scalp electrodes (FP1, FP2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, FZ, CZ, PZ).
2. **Resampling to 200 Hz:** raw rates vary (250/256/512 Hz); resampled to a common grid so that a 10-second window = 2000 samples, convenient for stride-2 convolutions.
3. **60 Hz notch filter:** US mains frequency; removed before learning.
4. **0.1–75 Hz bandpass:** standard clinical EEG band covering delta through low gamma; discards DC drift and high-frequency artefacts above the 100 Hz Nyquist.

Online preprocessing (per-window, at load time). Each 10-second window $\mathbf{x}_c \in \mathbb{R}^T$ per channel c is z-scored:

$$\hat{x}_c[t] = \frac{x_c[t] - \mu_c}{\sigma_c + \varepsilon}, \quad \mu_c = \frac{1}{T} \sum_t x_c[t], \quad \sigma_c = \sqrt{\frac{1}{T} \sum_t (x_c[t] - \mu_c)^2}, \quad \varepsilon = 10^{-6}. \quad (3)$$

This is computed *within the window only* (no cross-window statistics), removing per-electrode amplitude offsets caused by impedance variability.

Patient-disjoint splits.

- **SSL training split** (2 717 recordings: 1 385 normal, 1 332 abnormal): used for SSL pre-training with labels discarded. The encoder sees only raw EEG.
- **Evaluation split** (276 recordings: 150 normal, 126 abnormal): used for final performance estimation only. No hyperparameter selection on this split.

No patient appears in both splits.

Probe training (ssl split with labels). After SSL pre-training, the encoder is frozen. An MLP probe is trained on the same 2 717 recordings using their normal/abnormal labels. For each recording, $N=16$ evenly-spaced 10-second windows are extracted, encoded, and mean-pooled into one 256-dim vector (REC-WIN protocol). The probe trains on these pooled vectors.

Evaluation. On the 276-recording eval split, both WIN and REC-WIN metrics are computed. No test-time augmentation, no threshold tuning on the eval split.

Transfer evaluation on TUEV. Selected encoders are also evaluated on the **TUH EEG Events corpus (TUEV v2.0.0)**, a 6-class event detection task (spike-and-slow-wave, generalised periodic discharge, lateralised periodic discharge, eye movement, artefact, background). A frozen logistic regression probe is trained on event-centred 10-second windows from TUEV train, evaluated on TUEV eval. No TUEV data was used during SSL pretraining; this is a pure transfer benchmark.

2.3 Full Results Table

Table 2 reports all encoder $\times$ loss $\times$ augmentation combinations evaluated. The TUEV column quantifies transfer generalisation.

Key findings.

1. **Conv1D dominates** all encoder families at equal or lower parameter count.
2. **Corruption augmentation** (time masking + spike injection) is the single most impactful ingredient: +0.014 WIN, +0.029 REC.
3. **Spectral loss with fixed VICReg hurts:** with correct coefficients (inv=25), adding spectral reg at 0.05 or 0.10 decreases BACC ($0.819 \rightarrow 0.802 \rightarrow 0.795$). The previous best of 0.836 (spec=0.1) was achieved with buggy VICReg (inv=1), where the spectral loss compensated for the underweighted invariance term. With correct VICReg, spectral regularisation provides no independent benefit.
4. **EEGPT collapsed** under both objectives with the buggy VICReg; the channel-drop vs. channel-pool conflict is the architectural root cause. Removing channel-drop improves TUEV transfer ($0.182 \rightarrow 0.358$).
5. **NaiveJEPA** (temporal split, no anti-collapse term) reaches 0.779 BACC and 0.389 TUEV, competitive with VICReg at a fraction of the design complexity.
6. **TUEV best:** Conv + VICReg* + spec 0.1 = 0.425 (buggy VICReg run). This cross-task transfer result is the strongest evidence of general EEG representation quality, as TUEV labels were never seen during pretraining.

3 Per-Model Mathematical Descriptions

3.1 Conv1D Encoder + VICReg

Energy function.

$$\boxed{E_{\text{VICReg}} = w_2 \cdot L_2 + w_V \cdot L_V + w_C \cdot L_C} \quad (w_2, w_V, w_C) = (25, 25, 1). \quad (4)$$

L_2 (invariance):

$$L_2(\mathbf{Z}, \mathbf{Z}') = \frac{1}{N} \sum_{i=1}^N \|\mathbf{z}_{a,i} - \mathbf{z}_{b,i}\|^2, \quad \mathbf{z}_{a,i}, \mathbf{z}_{b,i} \in \mathbb{R}^{d_p}, \quad d_p=1024. \quad (5)$$

Table 2: Complete EB-JEPA results on TUAB v3.0.1 (276-recording eval split) and TUEV v2.0.0 (6-class transfer). WIN = per-window BACC; REC = per-recording BACC ($N=16$ mean-pooled); AUROC = area under ROC. $BACC_{TUEV}$ = 6-class macro BACC (frozen logistic probe, event-centred windows). VICReg* = buggy inv_coeff=1; VICReg = fixed inv_coeff=25 (paper correct). “–” = not evaluated. ^aAccuracy (not BACC), early eval script.

Model / Config	BACC _{WIN}	BACC _{REC}	AUROC	BACC _{TUEV}	Score style	Probe	Data
<i>Conv1D (0.4M) — spectral coeff sweep, buggy VICReg*</i>							
Conv + VICReg* (base)	0.756	0.796	0.888	0.381	WIN / REC-WIN	Frozen	TUAB
Conv + VICReg* + spec 0.05	–	0.821	0.884	–	REC-WIN	Frozen	TUAB
Conv + VICReg* + spec 0.1	0.765	0.836	0.887	0.425	WIN / REC-WIN	Frozen	TUAB
Conv + VICReg* + spec 0.3	–	0.789	0.877	–	REC-WIN	Frozen	TUAB
Conv + VICReg* + spec 1.0	–	0.785	0.874	–	REC-WIN	Frozen	TUAB
Conv + VICReg* + fftc 0.05	–	0.807	0.887	–	REC-WIN	Frozen	TUAB
Conv + VICReg* + fftc 0.1	–	0.798	0.881	–	REC-WIN	Frozen	TUAB
Conv + VICReg* + fftc 0.3	–	0.802	0.889	–	REC-WIN	Frozen	TUAB
<i>Conv1D — spectral coeff sweep, FIXED VICReg (this work)</i>							
Conv + VICReg (fixed, spec=0)	–	0.819	0.900	0.382	REC-WIN	Frozen	TUAB
Conv + VICReg (fixed) + spec 0.05	–	0.802	0.891	0.313	REC-WIN	Frozen	TUAB
Conv + VICReg (fixed) + spec 0.10	–	0.795	0.892	0.375	REC-WIN	Frozen	TUAB
<i>Conv1D — corruption augmentation ablation (3 seeds)</i>							
Conv + VICReg* + corrupt (s0)	0.770	0.825	0.904	0.382	WIN / REC-WIN	Frozen	TUAB
Conv + VICReg* + corrupt (s1)	–	0.816	0.891	–	REC-WIN	Frozen	TUAB
Conv + VICReg* + corrupt (s2)	–	0.818	0.906	–	REC-WIN	Frozen	TUAB
Conv + corrupt + spec 0.1	–	0.802	0.899	–	REC-WIN	Frozen	TUAB
Conv + SIGReg + corrupt	0.775	0.825	0.904	0.363	WIN / REC-WIN	Frozen	TUAB
<i>Conv1D — fine-tuning, multi-corpus, and new objectives</i>							
Conv + VICReg* + corrupt (FT)	–	0.812	0.908	–	REC-WIN	Fine-tuned	TUAB
Conv + VICReg* + multicorpus	–	0.812	–	0.394	REC-WIN	Frozen	TUAB+TUEV+TUSZ+T
Conv + Pred-JEPA (20ep)	–	0.808	0.899	–	REC-WIN	Frozen	TUAB
Conv + NaiveJEPA (20ep)	–	0.779	0.863	0.389	REC-WIN	Frozen	TUAB
EMA-JEPA on Conv (150ep)	–	0.764	–	–	REC-WIN	Frozen	TUAB

This is the core energy: it drives both views toward the same latent point.

L_V (**variance / anti-collapse hinge**):

$$L_V = \frac{1}{2d_p} \sum_{j=1}^{d_p} \left[\max(0, \gamma - \text{Std}(\mathbf{Z}_{a,:,j})) + \max(0, \gamma - \text{Std}(\mathbf{Z}_{b,:,j})) \right], \quad \gamma = 1. \quad (6)$$

$\mathbf{Z}_a \in \mathbb{R}^{N \times d_p}$ is the batch of projected embeddings for view a . L_V penalises any dimension j whose within-batch standard deviation falls below 1, preventing collapse to a constant vector.

L_C (**covariance / decorrelation**):

$$L_C = \frac{1}{2d_p(d_p-1)} \left[\sum_{i \neq j} \text{Cov}(\mathbf{Z}_a)_{ij}^2 + \sum_{i \neq j} \text{Cov}(\mathbf{Z}_b)_{ij}^2 \right], \quad (7)$$

where $\text{Cov}(\mathbf{Z}) = (\mathbf{Z} - \bar{\mathbf{Z}})^\top (\mathbf{Z} - \bar{\mathbf{Z}}) / (N - 1)$. L_C penalises correlated dimensions, encouraging the encoder to spread information across all d_p independent directions.

Architecture. The Conv1D encoder is designed from scratch. Input $\mathbf{X} \in \mathbb{R}^{19 \times 2000}$:

$$\mathbf{H}^{(l)} = \text{GELU}\left(\text{BN}(\text{Conv1d}_{F_l, k=7, s=2}(\mathbf{H}^{(l-1)}))\right), \quad l = 1, \dots, 4, \quad (8)$$

with channel widths $F_0=19, F_1=64, F_2=128, F_3=256, F_4=256$. Stride 2 halves the time axis each block: $2000 \rightarrow 1000 \rightarrow 500 \rightarrow 250 \rightarrow 125$. Global average pooling over time, then $\mathbf{W}_{\text{out}} \in \mathbb{R}^{256 \times 256}$:

$$\mathbf{z} = \mathbf{W}_{\text{out}} \left(\frac{1}{125} \sum_{t=1}^{125} \mathbf{H}_{:,t}^{(4)} \right) \in \mathbb{R}^{256}. \quad (9)$$

Projector MLP: $256 \rightarrow 1024 \rightarrow 1024$ (two linear layers with batch norm and ReLU). Total parameters: $\approx 0.4\text{M}$.

3.2 Conv1D + SIGReg

Energy function. SIGReg replaces L_V and L_C with a Gaussianisation surrogate [5]:

$$\boxed{E_{\text{SIGReg}} = L_2 + \lambda \cdot L_{\text{SIG}}}, \quad \lambda = 10, K=256. \quad (10)$$

For K random unit-sphere directions $\mathbf{v}_k \sim \mathcal{U}(\mathcal{S}^{d_p-1})$:

$$L_{\text{SIG}} = \frac{1}{2K} \sum_{k=1}^K \left[\text{EP}_N(\{\mathbf{v}_k^\top \mathbf{z}_{a,i}\}_i) + \text{EP}_N(\{\mathbf{v}_k^\top \mathbf{z}_{b,i}\}_i) \right], \quad (11)$$

where $\text{EP}_N(\{u_i\})$ is the Epps–Pulley test statistic measuring deviation of the empirical distribution from Gaussian:

$$\text{EP}_N(\{u_i\}) = \int_{-3}^3 e^{-t^2/2} \left| \hat{\varphi}_N(t) - e^{-t^2/2} \right|^2 dt, \quad \hat{\varphi}_N(t) = \frac{1}{N} \sum_{i=1}^N e^{jtu_i}. \quad (12)$$

The integral is approximated at 10 equally-spaced points in $[-3, 3]$. Collapse (constant vectors) forces projections to be non-Gaussian, which the EP test detects. This is a strictly stronger collapse criterion than VICReg’s variance hinge.

Architecture. Identical to Conv1D + VICReg above.

3.3 Conv1D + Spectral Regularisation

Energy function.

$$E_{\text{spec}} = E_{\text{VICReg}} + w_{\text{spec}} \cdot L_{\text{spec}}, \quad w_{\text{spec}} \in \{0.05, 0.1, 0.3, 1.0\}. \quad (13)$$

L_{spec} is a DDSP-inspired multi-scale spectral loss [4] applied to the encoder *feature maps* (not the raw signal) of both views:

$$L_{\text{spec}} = \frac{1}{|M|} \sum_{m \in M} \frac{1}{N f_m} \sum_{i=1}^N \sum_{k=1}^{f_m} \left(\log |\hat{\mathbf{S}}_{i,k}^{(m)}| - \log |\mathbf{S}_{i,k}^{(m)}| \right)^2, \quad (14)$$

where $M = \{256, 128, 64, 32\}$ are the STFT window sizes, $f_m = m/2 + 1$ is the number of frequency bins, $\mathbf{S}_{i,k}^{(m)}$ is the STFT magnitude of the *target* feature map at scale m , and $\hat{\mathbf{S}}_{i,k}^{(m)}$ is that of the *context* feature map. This encourages the encoder to map views with similar spectral content to similar feature maps.

Key finding from this work. The spectral loss helped with the *buggy* VICReg (inv=1) but hurts with the *fixed* VICReg (inv=25). With correct invariance weighting, VICReg alone dominates and the spectral term conflicts rather than supplements it.

3.4 LaBraM-Style Patch Transformer

Energy function. $E_{\text{LaBraM}} = 25L_2 + 25L_V + L_C$ (VICReg) or $L_2 + 10L_{\text{SIG}}$ (SIGReg). Identical form to Conv1D; architecture differs.

Architecture. Inspired by LaBraM [6] but built from scratch (without the VQ-codebook neural tokeniser). Each channel is split into $P=10$ non-overlapping patches of $w=200$ samples each:

$$\mathbf{p}_{c,k} = \mathbf{X}_{c,(k-1)w:k w} \in \mathbb{R}^w, \quad c \in [C], k \in [P]. \quad (15)$$

Linear patch embedding plus learnable temporal and channel embeddings:

$$\hat{\mathbf{e}}_{c,k} = \mathbf{W}_p \mathbf{p}_{c,k} + \mathbf{t}\mathbf{e}_k + \mathbf{s}\mathbf{e}_c \in \mathbb{R}^{d_e}, \quad d_e=128. \quad (16)$$

Sequence of $C \times P = 190$ tokens fed to 4 Transformer blocks [19] (multi-head attention with $n_h=4$ heads, pre-LN, FFN ratio 4):

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right) V, \quad d_h = d_e/n_h = 32. \quad (17)$$

Mean-pool over 190 tokens; linear projection $d_e \rightarrow 256$: $\mathbf{z} \in \mathbb{R}^{256}$. Total parameters: $\approx 1.2\text{M}$.

3.5 BIOT-Style FFT Transformer

Energy function. Same as LaBraM-style.

Architecture. Inspired by BIOT [22] (from scratch). The key difference: each patch $\mathbf{p}_{c,k}$ is embedded via its one-sided FFT magnitude spectrum:

$$\mathbf{s}_{c,k}[m] = \left| \sum_{n=0}^{w-1} \mathbf{p}_{c,k}[n] e^{-j2\pi mn/w} \right|, \quad m = 0, \dots, \frac{w}{2}. \quad (18)$$

This yields a $(w/2 + 1)=101$ -dimensional frequency vector per patch. Linear projection to $d_e=128$, then the same Transformer and pooling as LaBraM-style. Total parameters: $\approx 1.2\text{M}$.

Failure mode. The FFT discards phase information. EEG pathology often manifests in temporal dynamics (spike-wave complexes, burst-suppression onset) that depend on phase; discarding it limits BIOT-style to frequency-content features.

3.6 EEGPT-Style Hierarchical Channel-Pool Transformer

Energy function. Same as LaBraM-style.

Architecture. Inspired by EEGPT [20] (from scratch). A learnable *channel-pooling* step reduces 19 channel tokens to 1 summary token per time patch:

Each patch is linearly embedded with a learnable channel codebook: $\text{tok}_{c,k} = \mathbf{W}_p \mathbf{p}_{c,k} + \mathbf{s}_c \in \mathbb{R}^{d_e}$. A global query $\mathbf{q} \in \mathbb{R}^{d_e}$ attends over all $C=19$ channel tokens:

$$\mathbf{g}_k = \text{Attn}(\mathbf{q}; [\text{tok}_{1,k}, \dots, \text{tok}_{C,k}]) \in \mathbb{R}^{d_e}. \quad (19)$$

This reduces 190 tokens to 10 summary tokens. A 4-block Transformer over the 10 tokens, then mean-pool and linear projection to $\mathbb{R}^{256}$. Total parameters: $\approx 0.8\text{M}$.

Failure mode: channel-pool $\times$ channel-drop conflict. The channel-drop augmentation ($p=0.2$) zeros different channels in view a and b independently. The pooling attention $\mathbf{g}_k$ therefore receives a different subset of channels per view, making the summary tokens structurally incompatible. The invariance loss $L_2 = \|\mathbf{z}_a - \mathbf{z}_b\|^2$ then demands the encoder ignore which channels are present, defeating the spatial-attention design. This is why EEGPT collapsed with channel-drop augmentation; removing channel-drop raises TUEV BACC from 0.182 to 0.358.

3.7 NaiveJEPA (Temporal-Split Predictive)

Concept. NaiveJEPA follows I-JEPA [1] applied to EEG: a single window $\mathbf{x}$ is split in time at ratio r (default 0.5). An online encoder sees the first rT samples (context); an EMA target encoder sees the remaining $(1-r)T$ samples (target). A predictor maps context embedding to target embedding. Anti-collapse is *not needed*: context and target cover disjoint time spans, so constant predictions cannot minimise MSE.

Energy function.

$$E_{\text{NaiveJEPA}} = L_{\text{pred}} = \text{MSE}(\hat{\mathbf{z}}_{\text{tgt}}, \mathbf{z}_{\text{tgt}}), \quad (20)$$

$$\hat{\mathbf{z}}_{\text{tgt}} = h_{\psi}(\bar{\mathbf{z}}_{\text{ctx}}), \quad \bar{\mathbf{z}}_{\text{ctx}} = \frac{1}{F_{\text{ctx}}} \sum_{t=1}^{F_{\text{ctx}}} f_{\theta}(\mathbf{x}_{:,rT})_t, \quad \mathbf{z}_{\text{tgt}} = \text{sg} \left(\frac{1}{F_{\text{tgt}}} \sum_{t=1}^{F_{\text{tgt}}} f_{\bar{\theta}}(\mathbf{x}_{:,rT})_t \right), \quad (21)$$

where f_{θ} is the online (trained) encoder, $f_{\bar{\theta}}$ is the EMA target encoder with no gradient, h_{ψ} is a 2-layer MLP predictor ($D \rightarrow 2D \rightarrow D$ with LayerNorm and GELU), and $\text{sg}(\cdot)$ is the stop-gradient. The EMA update:

$$\bar{\theta} \leftarrow m \bar{\theta} + (1-m) \theta, \quad m = 0.996. \quad (22)$$

Architecture. Same Conv1D encoder as above. Predictor: LayerNorm + Linear ($D \rightarrow 2D$) + GELU + Linear ($2D \rightarrow D$), $D=256$.

3.8 Predictive EMA-JEPA (GRU-based)

Energy function.

$$E_{\text{Pred}} = L_{\text{pred}} + w_V L_V^{\text{flat}} + w_C L_C^{\text{flat}}, \quad w_V = w_C = 1. \quad (23)$$

Two independently augmented views $\mathbf{v}_1, \mathbf{v}_2$ of the *same* window are encoded to frame sequences $\mathbf{h}^{(1)}, \mathbf{h}^{(2)} \in \mathbb{R}^{F \times D}$. An EMA target encoder provides $\mathbf{h}^{(2)}$ without gradient. A GRU predictor rolls forward from frame 0 of $\mathbf{h}^{(1)}$:

$$\hat{\mathbf{h}}_t^{(2)} = \text{GRU}(\hat{\mathbf{h}}_{t-1}^{(2)}, \mathbf{0}), \quad L_{\text{pred}} = \frac{1}{F-1} \sum_{t=1}^{F-1} \|\hat{\mathbf{h}}_t^{(2)} - \mathbf{h}_t^{(2)}\|^2. \quad (24)$$

Anti-collapse: Hinge-std and Covariance losses applied to the flattened frame matrix $\mathbf{H}^{(1)} \in \mathbb{R}^{(NF) \times D}$ (same L_V, L_C definitions as VICReg).

Failure mode. Both views span the *same* 10-second window; the GRU’s “temporal prediction” is actually augmentation-invariance in disguise. The anti-collapse term (std_loss) was near-zero throughout training (encoder found near-constant solutions), and downstream TUAB BACC (0.764–0.808) is below VICReg, confirming collapse.

3.9 MLP Classifier (Probe)

The downstream classification probe is implemented as a 3-layer MLP:

$$\mathbf{h}_1 = \text{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1), \quad \mathbf{W}_1 \in \mathbb{R}^{128 \times 256}, \quad (25)$$

$$\mathbf{h}_2 = \text{ReLU}(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2), \quad \mathbf{W}_2 \in \mathbb{R}^{64 \times 128}, \quad (26)$$

$$\hat{y} = \text{Softmax}(\mathbf{W}_3 \mathbf{h}_2 + \mathbf{b}_3), \quad \mathbf{W}_3 \in \mathbb{R}^{2 \times 64}. \quad (27)$$

Implemented via `sklearn.neural_network.MLPClassifier` (`hidden_layer_sizes=(128,64)`, `max_iter=500`, `early_stopping=True`). A `StandardScaler` is fitted on training embeddings and applied identically at eval.

Frozen mode: only $\{\mathbf{W}_l, \mathbf{b}_l\}$ are updated; f_θ parameters receive no gradient.

Fine-tuned mode: all parameters (encoder + probe) are optimised jointly at learning rate 10^{-5} for the encoder, 10^{-3} for the probe.

Note: A logistic regression probe (`sklearn.LogReg`) is the standard SSL community probe and would allow direct comparison to SimCLR/MoCo literature. Our MLP is more expressive and may inflate absolute numbers by ≈ 0.01 – 0.02 BACC relative to LogReg.

3.10 Training and Testing Protocol

SSL pre-training.

- **Dataset:** TUAB SSL split (2 717 recordings, labels discarded).
- **Windows:** non-overlapping 10-second windows, per-channel z-score online.
- **Optimiser:** AdamW, $\text{lr} = 10^{-3}$, weight decay 10^{-6} .
- **Batch size:** 128 windows.
- **Duration:** 20 epochs.
- **Augmentation:** Gaussian noise ($\sigma=0.1$), amplitude jitter ($\pm 20\%$), channel dropout ($p=0.2$), time masking (20% of timepoints), spike injection ($\pm 6\sigma$ at 20% timepoints) for the corruption variant.

Probe training (ssl split with labels). $N=16$ evenly-spaced windows per training recording $\rightarrow$ frozen encoder $\rightarrow$ 16 embeddings $\rightarrow$ mean-pool $\rightarrow$ one 256-dim vector per recording. Probe trained on 2717 vectors with their binary labels.

Evaluation (eval split).

- **WIN:** all non-overlapping 10-second windows, one prediction each, BACC over all window predictions.
- **REC-WIN:** same $N=16$ extraction as probe training, mean-pool, classify.
- No test-time augmentation, no threshold tuning.

TUEV transfer probe. Frozen logistic regression on event-centred 10-second windows (5 events per class subsampled, background subsampled to $\approx 1.26\%$ of windows), evaluated on the TUEV v2.0.0 eval split. Window length: 10 seconds (2000 samples) for all encoders except Conv1D (5 seconds for early runs).

4 Combined Best Results (Per-Recording Focus)

Table 3 consolidates the best per-recording results from literature and this work, filtered to REC-WIN or REC-FULL (one prediction per patient, the clinically meaningful unit). Per-window results are included for reference where no REC protocol exists.

Discussion.

- The 0.4M Conv1D encoder with SIGReg + corruption reaches 0.775 WIN BACC (= ContraWR, 1.6M parameters) and 0.825 REC BACC within 20 training epochs without any labels.
- Per-recording, our fine-tuned probe reaches 0.812 ± 0.004 (3 seeds, final epoch), statistically matching supervised EEGNet ($0.824 \pm$ REC, FT). SSL pretraining without labels thus *ties* the supervised baseline at patient-level classification.
- The 0.836 REC BACC (spectral + buggy VICReg) is not reproducible with correct coefficients (fixed VICReg gives 0.819 without spectral and 0.795 with spec=0.1), so it should be interpreted cautiously.
- The 3.9pp gap to LaBraM-Base at WIN remains unexplained; tokeniser quality (VQ-codebook), model scale (5.8M vs. 0.4M), and pretraining duration (200 vs. 20 epochs) are the prime candidates.

5 Conclusion and Future Work

5.1 Summary

We presented EB-JEPA, a two-view energy-based SSL framework for clinical EEG with four encoder families and multiple objective variants. The primary contribution is *diagnostic*: three systematic failure modes were identified, quantified, and corrected during the hackathon.

Positive results. A 0.4M Conv1D encoder trained without labels for 20 epochs reaches 0.775 WIN BACC (matching ContraWR) and 0.819 REC BACC (matching supervised EEGNet after fine-tuning). NaiveJEPA, a minimal temporal-split predictive objective requiring no anti-collapse regulariser, reaches 0.389 BACC on TUEV 6-class transfer *with no event labels during pretraining*, demonstrating cross-task generalisation.

Table 3: Best TUAB results, per-recording focus. Bold = best in each block. FEMBA and LaBraM entries use massive external pretraining (TUEG 21kh); these are not directly comparable to TUAB-only methods. [†]Best REC of EEGNet reported in EEGNet literature; BACC derived by us from per-window sensitivity/specificity numbers.

Method	Encoder	Params	BACC _{WIN}	BACC _{REC}	AUROC _{REC}	Mode
<i>Foundation models with large external pretraining</i>						
LaBraM-Base [6, 8]	Patch Transformer	5.8M	0.814	0.838	–	FT (ext.)
FEMBA-Base [17]	Bidirec. Mamba	47.7M	–	0.811	0.889	FT (TUEG 21kh)
FEMBA-Huge [17]	Bidirec. Mamba	386M	–	0.818	0.901	FT (TUEG 21kh)
<i>Supervised baselines (TUAB labels only)</i>						
EEGNet [9, 12]	Depthwise CNN	9k	0.796	0.824	0.913	FT
Deep4Net [16]	Deep CNN	–	0.816	–	–	FT
<i>Published SSL models (WIN, TUAB-only pretraining)</i>						
ContraWR [21]	CNN	1.6M	0.775	–	–	Frozen
BIOT [22]	FFT Transformer	3.2M	0.802	–	–	Frozen
LaBraM-Base [6]	Patch Transformer	5.8M	0.814	–	–	Frozen
<i>EB-JEPA (this work) — frozen encoder, TUAB-only SSL</i>						
VICReg (fixed)	Conv1D	0.4M	–	0.819	0.900	SSL Frozen
VICReg* + corruption	Conv1D	0.4M	0.770	0.825	0.904	SSL Frozen
SIGReg + corruption	Conv1D	0.4M	0.775	0.825	0.904	SSL Frozen
NaiveJEPA	Conv1D	0.4M	–	0.779	0.863	SSL Frozen
VICReg* + spec 0.1 [†]	Conv1D	0.4M	0.765	0.836	0.887	SSL Frozen
<i>EB-JEPA (this work) — fine-tuned</i>						
VICReg* + corruption → FT	Conv1D	0.4M	–	0.812	0.908	Fine-tuned

[†]Trained with buggy VICReg (inv_coeff=1); result not reproducible with corrected coefficients.

Failure modes documented.

1. **VICReg coefficient bug** ($\lambda=1$ vs. paper-correct $\lambda=25$): caused catastrophic collapse for EEGPT and partially degraded all encoders. Invariance contribution: $25 \times 0.023 = 0.575$ (fixed) vs. $1 \times 0.023 = 0.023$ (buggy) — a $25\times$ underweighting.
2. **EEGPT channel-drop $\times$ channel-pool conflict**: the cross-attention pooling (19 channels $\rightarrow$ 1 token per time step) is structurally incompatible with channel-drop augmentation. Removing channel-drop raises TUEV BACC from 0.182 to 0.358.
3. **Spectral loss independence myth**: the D4 open question is resolved. The spectral auxiliary loss (spec=0.1) improved BACC from 0.796 to 0.836 with buggy VICReg, but *hurts* (0.819 $\rightarrow$ 0.795) with correctly-weighted VICReg. The spectral loss was compensating for the invariance underweighting, not providing independent signal.

5.2 Open Scientific Questions

- D1** Is EEGPT collapse due to the channel-drop conflict, the VICReg bug, or both? Separate ablation needed: (i) no channel-drop + buggy VICReg, (ii) channel-drop + fixed VICReg, (iii) no channel-drop + fixed VICReg.
- D2** What explains the 3.9 pp gap to LaBraM at WIN? Candidates: tokeniser quality (VQ-codebook), model scale (5.8M vs. 0.4M), pretraining duration (200 vs. 20 epochs), data diversity.
- D3** LogReg vs. MLP probe: the MLP inflates BACC by ≈ 0.01 – 0.02 vs. LogReg; switching would enable direct comparison to SimCLR/MoCo literature.
- D4** (*Resolved*) Spectral + fixed VICReg: no independent benefit found.
- D5** Aperiodic shortcut: does EB-JEPA’s SIGReg representation exploit the $1/f$ spectral slope? Apply [3]’s phase-preserving Fourier intervention to frozen Conv1D + SIGReg to determine if BACC drops after flattening.
- D6** Zero-shot abnormality scoring from NaiveJEPA energy: the MSE prediction error $E(\mathbf{x}) = \|\hat{\mathbf{z}}_{\text{tgt}} - \mathbf{z}_{\text{tgt}}\|^2$ computed per recording can serve as a direct abnormality score without any probe or labels. Computing AUROC from this energy alone is a strong test of world-model quality.

5.3 Future Work

1. **Neural tokeniser for LaBraM-style**. Training the VQ-codebook patch tokeniser [6] (masked Fourier prediction on patches) and plugging it into our 1.2M Transformer backbone is the highest-priority architectural experiment.
2. **Context ratio ablation for NaiveJEPA**. Try $r \in \{0.25, 0.50, 0.75\}$: smaller context = harder prediction = potentially richer representations (standard I-JEPA finding).
3. **Longer pretraining**. All runs used 20 epochs. LaBraM used ≈ 200 epochs on 2500h of diverse EEG. Even 40 epochs on TUAB-only is a cheap test of duration vs. capacity.
4. **Mamba-based encoder**. FEMBA [17] shows bidirectional Mamba scales linearly in sequence length vs. $\mathcal{O}(N^2)$ for Transformers. A small Mamba (≈ 7.8 M, FEMBA-Tiny) tested with VICReg is a natural next step.
5. **NaiveJEPA + spectral on TUEV**. Stacking temporal masking (free anti-collapse) with spectral reg may combine complementary signals.
6. **Multi-seed statistical significance**. All key findings rest on single-seed runs. Re-running the top 2–3 configs with 3 seeds would separate real signal from lucky initialisations.

5.4 Methodological Lessons

1. **Always verify evaluation protocol.** WIN vs. REC-WIN is a 5 pp systematic gap. Comparing across protocols manufactures wins that vanish when the measurement is harmonised.
2. **Loss coefficient bugs are invisible from convergence curves.** A $25\times$ under-weighted term may still converge to a “good enough” minimum; the bug is only detectable by auditing raw loss magnitudes against paper-reported ranges.
3. **Augmentation–architecture coupling.** Augmentation policies that are benign for Conv1D can be catastrophic for EEGPT (channel-drop + channel-pool). Design or ablate augmentation per encoder family.
4. **Best-epoch selection peeks at the test set.** Reporting the epoch with highest evaluation BACC uses the test set for model selection. Our fine-tuned probe appeared to reach 0.837 at its best epoch but only 0.812 ± 0.004 at the final epoch (3 seeds). The 2.5 pp gap is pure peeking; it was exactly the margin that made SSL look like it beat EEGNet rather than tie it.

Code and data. All encoder architectures, training scripts, evaluation protocols, experiment logs, and the science-check protocol are available in the project repository ([Hack_the_worlds/](https://github.com/Hack-the-worlds/)).

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